

Chemistry — Electrochemistry (Chapter 7)

Board of Intermediate and Secondary Education | SSC-I (9th Grade) | Annual Examination 2026

Student: Seemab **Exam:** exam-04-ch7 **Attempt date:** 12 June 2026 **Marked by:** Automated marking system

TOTAL SCORE
60 / 65

PERCENTAGE
92.3%

GRADE
A+

MCQ
12 / 12

SEC B
29 / 33

SEC C
19 / 20

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks) Score: 12/12

#	Question (summary)	Seemab's Answer	Correct Answer	Result
1	In which change is nitrogen reduced?	C — N ₂ to NH ₃	C — N ₂ to NH ₃	✓
2	Which is an example of oxidation?	B — Ag to Ag ⁺	B — Ag to Ag ⁺	✓
3	In ZnO + H ₂ → Zn + H ₂ O, which is reduced?	B — ZnO	B — ZnO	✓
4	Oxidation state of Cr in K ₂ Cr ₂ O ₇	B — +6	B — +6	✓
5	Oxidation state of S in SO ₄ ²⁻	C — +6	C — +6	✓
6	H ₂ S acts as in H ₂ S + Cl ₂ → 2HCl + S	A — Reducing agent	A — Reducing agent	✓
7	An oxidizing agent is a substance that:	B — gains electrons	B — gains electrons	✓
8	Formula of iron(III) chloride (IUPAC)	C — FeCl ₃	C — FeCl ₃	✓
9	Rusting of iron requires:	C — oxygen and water	C — oxygen and water	✓
10	Cathodic protection of ships uses blocks of:	D — magnesium	D — magnesium	✓
11	Acidified KMnO ₄ changes to _____ when reducing agent added	B — colourless	B — colourless	✓
12	Oxidation number of free element is always:	C — zero	C — zero	✓

✓ **Perfect MCQ score — 12/12. This is Seemab's 13th perfect MCQ score in Chemistry across 4 Chemistry exams and the third subject to return 12/12 after the Math-07 MCQ break.**

SECTION B — Short Questions (11 × 3 = 33 Marks) Score: 29/33

Q.2(i)

OR variant: Oxidation and reduction in terms of hydrogen transfer, with example

2.5
/ 3

Definitions correct and clearly stated: "Oxidation is the loss of hydrogen and reduction is the gain of hydrogen." The acetylene combustion equation $2\text{C}_2\text{H}_2 + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$ is accurate and C₂H₂ losing hydrogen is identified as oxidation. The reduction side (O₂ gaining hydrogen to form H₂O) was not explicitly labelled in words — the third mark requires naming both sides of the example.

-0.5: Reduction identification not explicitly stated for the example. The question requires one example with both sides (oxidized species AND reduced species) named. Only the oxidized side was labelled.

Q.2(ii)

Main: 4 rules for assigning oxidation states + N in HNO₃

3
/ 3

All four rules stated precisely with good examples (S₈, H₂, N₂ for Rule 1; Na⁺, Ca²⁺ for Rule 2; CO₃²⁻ and NH₄⁺ for Rule 3; HCl and CO₂ for Rule 4). Working for N in HNO₃: (1)+x+3(-2)=0 → x=+5, fully correct.

Full marks. Excellent structured answer with textbook-precision examples for every rule.

Q.2(iii)**3**
/ 3Main: Oxidation state of Cr in $\text{K}_2\text{Cr}_2\text{O}_7$

Step-by-step working: $1(2) + x(2) + 7(-2) = 0 \rightarrow 2 + 2x - 14 = 0 \rightarrow 2x = 12 \rightarrow x = +6$. All three mark points covered (K contribution, O contribution, Cr calculation).

Full marks. Clean algebraic layout.

Q.2(iv)**3**
/ 3Main: Oxidation state of N in (a) NO_2 (b) N_2O (c) HNO_3

(a) NO_2 : $x + 2(-2) = 0 \rightarrow x = +4$ ✓. (b) N_2O : $2x + (-2) = 0 \rightarrow x = +1$ ✓ (correctly handled the 2N by dividing). (c) HNO_3 : $1 + x + 3(-2) = 0 \rightarrow x = +5$ ✓.

Full marks. All three working chains are clean and correct.

Q.2(v)**2.5**
/ 3

Main: Formulae of (a) copper(II) sulphate (b) iron(II) chloride (c) iron(III) chloride — state oxidation state of metal

Formulae are all correct: CuSO_4 , FeCl_2 , FeCl_3 . However the question explicitly asks to "state the oxidation state of the metal in each" — the oxidation states $\text{Cu} = +2$, $\text{Fe} = +2$, $\text{Fe} = +3$ were not written alongside the formulae. The names were copied, which implies the states, but the examiner expects an explicit statement.

-0.5: Oxidation states of the metals ($\text{Cu} = +2$, $\text{Fe(II)} = +2$, $\text{Fe(III)} = +3$) not explicitly written next to the formulae. The question says "state the oxidation state" as a separate requirement.

Q.2(vi)**3**
/ 3OR variant: Identify oxidizing and reducing agents in $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$

Oxidation state assignments shown clearly: Na: $0 \rightarrow +1$ (oxidized, reducing agent); Cl_2 : $0 \rightarrow -1$ (reduced, oxidizing agent). All three mark points earned: state changes assigned, Na identified as reducing agent, Cl_2 identified as oxidizing agent with justification.

Full marks. Efficient, textbook-style layout.

Q.2(vii)**2**
/ 3Main: In $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$, identify (a) element oxidized (b) element reduced (c) oxidizing agent (d) reducing agent

The chemistry is fully correct — oxidation states assigned accurately for all species (Fe : $+3 \rightarrow 0$ reduction; C : $+2 \rightarrow +4$ oxidation). However the sub-part labelling is inverted throughout: (a) was labelled "element reduced" and answered with $\text{Fe}_2\text{O}_3(\text{Fe})$: $+3 \rightarrow 0$ — but the question asks (a) for element oxidized. Sub-parts (a) and (b) were swapped, and (c) and (d) were similarly swapped. The correct content exists in the answer but is filed under the wrong letter headings.

-1: Sub-part labels (a)/(b) swapped and (c)/(d) swapped throughout. Under exam conditions, answers filed under wrong sub-headings lose the mark even when the underlying chemistry is right. Always re-read the sub-part letters before writing.

Q.2(viii)**3**
/ 3OR variant: Activity 7.2 — H_2O_2 added to KI solution

Observation correct (solution turns reddish-brown, formation of iodine I_2). Equation $\text{H}_2\text{O}_2 + 2\text{KI} + 2\text{H}^+ \rightarrow 2\text{K}^+ + \text{I}_2 + 2\text{H}_2\text{O}$ with oxidation states shown. Oxidizing agent H_2O_2 (O: $-1 \rightarrow -2$, reduced) and reducing agent KI (I: $-1 \rightarrow 0$, oxidized) both correctly identified.

Full marks. Activity 7.2 answered with equation, observation, and both agent identifications.

Q.2(ix)**3**
/ 3

Main: Define corrosion + two half-reactions for rusting

Definition is precise and complete: "natural electrochemical process in which a metal reacts with oxygen and moisture to form rust, involving simultaneous oxidation (anodic region) and reduction (cathodic region) with electrons flowing through the metal." Anodic reaction: $\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$. Cathodic reaction: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$. Both labelled correctly.

Full marks. All three components present with correct region labels and state symbols.

Q.2(x)**2**
/ 3

Main: Galvanizing and electroplating as corrosion prevention methods

Galvanizing: principle correct (zinc acts as physical barrier and sacrificial anode) but the process description is absent — the textbook specifies dipping in zinc chloride bath, then zinc bath, then cooling. Electroplating: principle correct (electrolytic process deposits corrosion-resistant metal) but the electrode setup is not described (object = cathode, plating metal = anode, electrolyte solution). Both answers capture the "what" but not the "how."

-1 total: Galvanizing process steps missing (-0.5); electroplating electrode arrangement not described (-0.5). The marking scheme awards 1.5 marks per method, with 0.5 requiring the process detail.

Q.2(xi)**2**
/ 3Main: Oxidation state of central atom in (a) SO_4^{2-} (b) PO_4^{3-} (c) NH_4^+

(a) SO_4^{2-} : $x + 4(-2) = -2 \rightarrow x = +6$ ✓. (b) PO_4^{3-} : $x + 4(-2) = -3 \rightarrow x = +5$ ✓. (c) NH_4^+ : $x + 4(1) = +1 \rightarrow x + 4 = 1 \rightarrow x = 1 - 4$. The arithmetic at the final step produced $x = -5$ instead of $x = -3$. The working was correct up to " $x = 1 - 4$ " but $1 - 4 = -3$, not -5.

-1 (Q.2(xi)(c)): Arithmetic error in NH_4^+ — wrote $x = 1 - 4 = -5$ but $1 - 4 = -3$. Method is correct; pure subtraction slip. N in $\text{NH}_4^+ = -3$. This is the same calculation that appeared in Q.3(i)(or)(ii) where it was done correctly ($N = -3$ in Section C) — suggesting a moment of inattention rather than a concept gap.

SECTION C — Long Questions (4 × 5 = 20 Marks) Score: 19/20**Q.3(i)****5**
/ 5OR variant: Four rules for assigning oxidation states + calculations for B in H_3BO_3 , N in NH_4^+ , Mn in KMnO_4

(a) All four rules stated with excellent examples — Rule 1: S_8 , H_2 , N_2 , I_2 ; Rule 2: $\text{Na}^+ = +1$, $\text{Ca}^{2+} = +2$; Rule 3: CO_3^{2-} (sum = -2) and NH_4^+ (sum = +1) both given as examples; Rule 4: HCl (sum = 0) and CO_2 (sum = 0). Rule 3 example is the carbonate ion from Example 7.5, exactly as the key requires. (b) B in H_3BO_3 : $1(3) + x + 3(-2) = 0 \rightarrow x = +3$ ✓. N in NH_4^+ : $x + 4(1) = +1 \rightarrow x = -3$ ✓ (correct here). Mn in KMnO_4 : $1 + x + 4(-2) = 0 \rightarrow x = +7$ ✓.

Full marks — 5/5. Notably, the NH_4^+ calculation that caused the -1 in Q.2(xi) was done correctly here, confirming it was a momentary slip rather than a knowledge gap.

Q.3(ii)**4**
/ 5

Main: Define oxidizing and reducing agents + explain why each is itself the opposite; identify agents in 3 reactions

(a) Definitions: The oxidizing agent definition is essentially correct ("takes electrons from it, electron acceptor") but the clarifying note in parentheses reads "(they lose electrons)" which is the wrong direction — oxidizing agents gain electrons. Similarly the reducing agent parenthetical states "(they gain electrons)" but reducing agents lose electrons when they donate them. The core definitions are right but the parenthetical notes contradict them. (b) Three reactions: all fully correct with oxidation state assignments and agent identifications for $2\text{S} + \text{Cl}_2 \rightarrow \text{S}_2\text{Cl}_2$ (S: $0 \rightarrow +2$ oxidized; Cl: $0 \rightarrow -1$ reduced), $2\text{Na} + \text{Br}_2 \rightarrow 2\text{NaBr}$ (Na: $0 \rightarrow +1$ oxidized; Br: $0 \rightarrow -1$ reduced), $\text{Mg} + \text{H}_2\text{O} \rightarrow \text{MgO} + \text{H}_2$ (Mg: $0 \rightarrow +2$ oxidized; H: $+1 \rightarrow 0$ reduced).

-1: In part (a), the bracketed clarifications contain a conceptual reversal. "Oxidizing agents are reduced in the process (they lose electrons)" — they are reduced precisely because they GAIN electrons, not lose them. And "reducing agents are oxidized in the process (they gain electrons)" — they are oxidized because they LOSE electrons. These parenthetical errors in a definition-focused sub-part cost 0.5 per definition.

Q.3(iii)**5**

OR variant: Roman numerals in naming + oxidation state of underlined element in 4 compounds

/ 5

(a) Roman numerals: "used to indicate oxidation states of elements in compounds when metal exhibits variable oxidations in compounds." Three textbook examples: CuSO_4 = copper(II)sulphate, FeCl_2 = iron(II)chloride, FeCl_3 = iron(III)chloride. All correct. (b) Calculations: N in NO_2 : $x + 2(-2) = 0 \rightarrow x = +4$ ✓. S in H_2SO_4 : $2(1) + x + 4(-2) = 0 \rightarrow x = +6$ ✓. Cr in Cr_2O_3 : $x(2) + 3(-2) = 0 \rightarrow 2x = 6 \rightarrow x = +3$ ✓. Mn in MnO_4^- : $x + 4(-2) = -1 \rightarrow x - 8 = -1 \rightarrow x = +7$ ✓.

Full marks — 5/5. All four oxidation state calculations correct with full step-by-step working. The H_2SO_4 working used 1(2) for 2H_1 correctly.

Q.3(iv)**5**

Main: Corrosion — definition, electrochemical nature, half-reactions + four prevention methods

/ 5

(a) Definition matches the answer key precisely: natural electrochemical process, metal reacts with oxygen and moisture, simultaneous oxidation (anodic region) and reduction (cathodic region), electrons flow through metal. Anodic: $\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$. Cathodic: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$. Fe^{2+} flow to cathodic region noted. (b) All four methods present with correct principles: paint (physical barrier, cheap), alloying (chromium forms self-healing oxide layer), galvanizing (physical barrier + sacrificial anode), electroplating (electrolytic process, deposit corrosion-resistant metal). Note: "self healing" was written for alloying — the key says "self-healing oxide layer" which matches.

Full marks — 5/5. Outstanding answer: electrochemical nature explained with mechanism, both half-reactions with state symbols and labels, all four prevention methods with distinct principles. The corrosion topic was mastered completely in this exam.

Score Summary

Section	Max Marks	Marks Awarded	Percentage	Status
A — MCQs (12 questions)	12	12	100%	PERFECT
B — Short Questions (11 questions)	33	29	87.9%	4 marks dropped
C — Long Questions (4 questions)	20	19	95.0%	1 mark dropped
TOTAL	65	60	92.3%	A+

Deductions at a Glance

Question	Marks Lost	Reason
Q.2(i)(or)	-0.5	Reduction identification not explicitly named for the hydrogen-transfer example
Q.2(v)	-0.5	Oxidation states of metals not explicitly stated alongside formulae (question specifically says "state the oxidation state")
Q.2(vii)	-1	Sub-part labels (a)/(b) and (c)/(d) inverted — correct chemistry filed under wrong letter headings
Q.2(x)	-1	Galvanizing process steps (-0.5) and electroplating electrode arrangement (-0.5) absent — principle stated but not the mechanism
Q.2(xi)(c)	-1	Arithmetic slip: $x = 1 - 4$ written as -5 instead of -3 (N in NH_4^+)
Q.3(ii)(a)	-1	Parenthetical notes on oxidizing/reducing agents contradict the definitions — "they lose electrons" for oxidizing agent and "they gain electrons" for reducing agent are both inverted
TOTAL DROPPED	-5	

Strengths and Areas for Revision**Strengths in this exam**

- Perfect MCQ — 12/12 including tricky Q10 (magnesium)

Areas for revision

- **Sub-part label discipline:** Q.2(vii) had correct content

not zinc for ships) and Q11 (colourless, not blue)

- Oxidation state calculations — flawless across all five compounds in 3(iii)(b) and all three in 2(iv)
- Corrosion mastery — identical answer in both Q.2(ix) and Q.3(iv); both half-reactions written with correct state symbols and region labels
- Activity 7.2 ($\text{H}_2\text{O}_2 + \text{KI}$) answered with full equation, oxidation states and both agent identifications
- Four rules for oxidation states cited with multiple textbook examples per rule in Q.3(i)(or)
- Section C long answers are structured with headings and sub-numbering — consistently professional layout
- All 11 Section B questions attempted — no omissions, no rushed endings (a historical issue now firmly resolved)

in wrong slots — costs 1 mark. Always re-read (a)(b)(c)(d) before writing each sub-answer.

- **Electron direction for agents (new flag):** Oxidizing agents GAIN electrons (they are reduced). Reducing agents LOSE electrons (they are oxidized). The parenthetical notes in Q.3(ii)(a) had both reversed — a conceptual confusion that needs drilling.
- **Process details for methods:** Q.2(x) galvanizing and electroplating — know the steps: ZnCl_2 bath $\rightarrow$ zinc bath for galvanizing; cathode = object, anode = plating metal for electroplating.
- **Arithmetic under pressure:** Q.2(xi)(c) — $1-4=-3$ not -5 . The same calculation was done correctly in Q.3(i). Slow down on final arithmetic line.
- **State the oxidation state explicitly:** Q.2(v) — writing just the formula is not enough; the question says "state the oxidation state of the metal in each." Write $\text{Cu}=+2$, $\text{Fe(II)}=+2$, $\text{Fe(III)}=+3$ explicitly.

Chemistry Progress: All Attempts Compared

Exam	Date	Chapters	MCQ	Sec B	Sec C	Total	%	Grade
exam-01-ch1-2	12 Apr 2026	Ch 1-2	12/12	31/33	21/20*	64/65	98.5%	A+
exam-02-ch3-5	17 Apr 2026	Ch 3-5	12/12	28/33	19.5/20	59.5/65	91.5%	A+
exam-03-ch6	12 May 2026	Ch 6	12/12	27.5/33	18/20	57.5/65	88.5%	A+
exam-04-ch7 ►	12 Jun 2026	Ch 7	12/12	29/33	19/20	60/65	92.3%	A+

* exam-01 bonus mark in Section C due to additional correct content awarded.

Trend: After dipping to 88.5% on the calculation-heavy stoichiometry exam (Ch6), Seemab rebounds to 92.3% on this exam. Section B improved by 1.5 marks and Section C by 1 mark compared to the previous attempt. Chemistry 4-exam average rises from 92.8% to **92.7%** (4 exams: $64+59.5+57.5+60 = 241/260$).